Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In forced oscillation of a particle, the amplitude is maximum for an angular frequency ω_1 of the applied force, while the energy is maximum for an angular frequency ω_2 of the applied force, then

(A) $\omega_1 = \omega_2$

(B) $\omega_1 > \omega_2$

(C) $\omega_1 < \omega_2$

(D) $\omega_1 < \omega_2$, when damping is small and $\omega_1 > \omega_2$ when damping is large.

Q.2

Two waves travelling in a medium are given by
$y_1 = 2\sin\left(3\pi t - \frac{\pi}{2}x\right)$
and $\quad y_2 = 4\sin\left(6\pi t - \frac{3\pi}{2}x\right)$
where y_1, y_2 and x are in cm and t in second. The displacement of a particle at $x = 1\ \mathrm{cm}$ at time $t = 1\ \mathrm{s}$ is

(A) $-2\ \mathrm{cm}$

(B) $-3\ \mathrm{cm}$

(C) $+2\ \mathrm{cm}$

(D) $+6\ \mathrm{cm}$

Q.3

Two waves each of amplitude $2\ \mathrm{cm}$ and wavelength $5\ \mathrm{cm}$ and frequency $10\ \mathrm{Hz}$ have a constant phase difference of 60°. Travelling in the same direction, they superpose at a particle of the medium. What is the resultant amplitude of the oscillations of the particle?

(A) $3\sqrt{3}\ \mathrm{cm}$

(B) $2\sqrt{3}\ \mathrm{cm}$

(C) $4\sqrt{3}\ \mathrm{cm}$

(D) $5\sqrt{3}\ \mathrm{cm}$

Q.4

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What is the period of a pendulum formed by pivoting a metre stick so that it is free to rotate about a horizontal axis passing through the $75\ \text{cm}$ mark?

(A) 153 s

(B) 125 s

(C) 134 s

(D) 324 s

Q.5

A simple pendulum of length $40\ \text{cm}$ oscillates with an angular amplitude of $0.04\ \text{rad}$. Find (a) the time period, (b) the linear amplitude of the bob, (c) the speed of the bob when the string makes 0.02 rad with the vertical and (d) the angular acceleration when the bob is in momentary rest. Take $g = 10\ \text{m/s}^2$.

(A) 1.26 s, 1.6 cm,6.8 cm/s, 1 rad/s^2

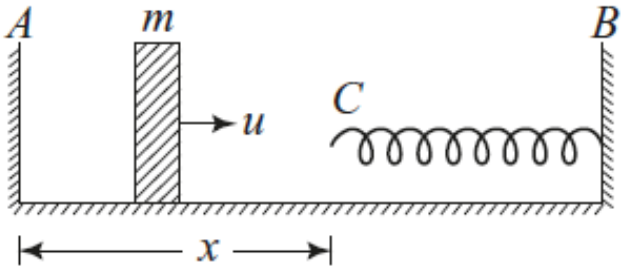
(B) 1.28 s, 1.6 cm,6.9 cm/s, 2 rad/s^2

(C) 1.3 s, 1.8 cm,6.9 cm/s, 1 rad/s^2

(D) 1.26 s, 1.7 cm, 7 cm/s, 2 rad/s^2

Q.6

Figure shows a block of mass $m = 2.5\ \text{kg}$ moving towards a spring of spring constant $k = 10\text{Nm}^{-1}$ with a constant velocity $u = 1\ \text{ms}^{-1}$.A and B are two rigid vertical walls. The block starts moving from A and strikes the end C of the spring at a distance $x = 50\ \text{cm}$. If the collision between the block and wall A is perfectly elastic, the time period T (in seconds) of the motion of the block will be



(A) $\left(\frac{\pi}{2}+1\right)$

(B) $(\pi+1)$

(C) $(2\pi+1)$

(D) 2π

Q.7

If a simple pendulum of length l has maximum angular displacement θ, then the maximum K.E. of bob of mass m is

(A) $\frac{1}{2}\text{ml/g}$

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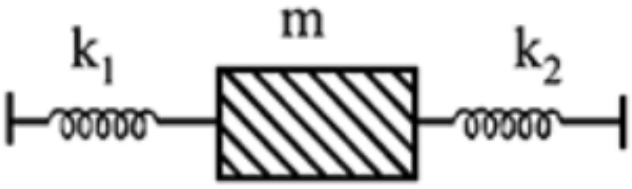
(B) $\mathrm{mg}/2l$

(C) $\mathrm{mg}\,l(1-\cos\theta)$

(D) $\mathrm{mg}l\sin\theta/2$

Q.8

Two springs, of force constants k_1 and k_2 are connected to a mass m as shown. The frequency of oscillation of the mass is f. If both k_1 and k_2 are made four times their original values, the frequency of oscillation becomes



(A) $2f$

(B) $f/2$

(C) $f/4$

(D) $4f$

Q.9

The bob of a simple pendulum executes simple harmonic motion in water with a period t, while the period of oscillation of the bob is t_0 in air. Neglecting frictional force of water and given that the density of the bob is $(4/3)\times 1000\ \mathrm{kg/m^3}$. The relationship between t and t_0 is

(A) $t=2t_0$

(B) $t=t_0/2$

(C) $t=t_0$

(D) $t=4t_0$

Q.10

A pendulum made of a uniform wire of cross sectional area A has time period T. When an additional mass M is added to its bob, the time period changes to $\mathrm{T_M}$. If the Young's modulus of the material of the wire is Y then $\frac{1}{Y}$ is equal to:
$(g=$ gravitational acceleration $)$

(A) $\left[1-\left(\frac{\mathrm{T_M}}{\mathrm{T}}\right)^2\right]\frac{\mathrm{A}}{\mathrm{Mg}}$

(B) $\left[1-\left(\frac{T}{T_M}\right)^2\right]\frac{A}{Mg}$

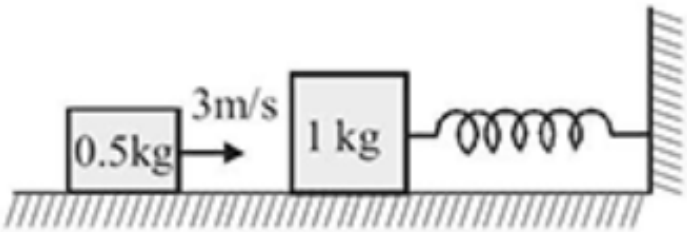
(C) $\left[\left(\frac{\mathrm{T_M}}{\mathrm{T}}\right)^2-1\right]\frac{\mathrm{A}}{\mathrm{Mg}}$

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(D) $\left[\left(\frac{T_M}{T}\right)^2 - 1\right] \frac{Mg}{A}$

Q.11

A $1\ \text{kg}$ mass is attached to a spring of force constant 600 N/m and rests on a smooth horizontal surface with other end of the spring tied to wall as shown in figure. A second mass of $0.5\ \text{kg}$ slides along the surface towards the first at $3\ \text{m/s}$. If the masses make a perfectly inelastic collision, then find amplitude and time period of oscillation of combined mass.



(A) $5\ \text{cm}, \frac{\pi}{10}\ \text{s}$

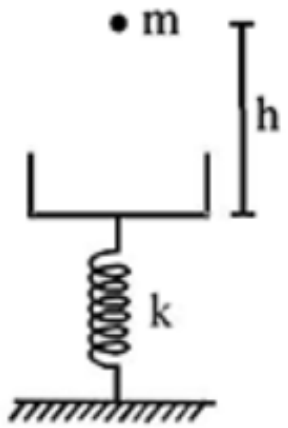
(B) $5\ \text{cm}, \frac{\pi}{5}\ \text{s}$

(C) $4\ \text{cm}, \frac{2\pi}{5}\ \text{s}$

(D) $4\ \text{cm}, \frac{\pi}{3}\ \text{s}$

Q.12

A mass m fall on spring of spring constant k and negligible mass from a height h. Assuming it sticks to the pan and executes simple harmonic motion, the maximum height upto which the pan will rise is



(A) $\frac{mg}{k}$

(B) $\frac{mg}{k}\left[\sqrt{1+\frac{2kh}{mg}} - 1\right]$

(C) $\frac{mg}{k}\left[\sqrt{1+\frac{2kh}{mg}} + 1\right]$

(D) $\frac{mg}{k}\left[\sqrt{1+\frac{kh}{mg}} - 1\right]$

Q.13

A simple pendulum attached to the ceiling of a stationary lift has a time period T. The distance y covered by

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the lift moving upwards varies with time t as $y = t^2$ where y is in metres and t in seconds. If $g = 10\ \mathrm{m/s^2}$, the time period of pendulum will be

(A) $\sqrt{\frac{4}{5}}\ \mathrm{T}$

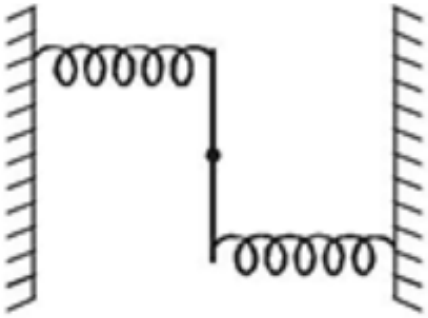
(B) $\sqrt{\frac{5}{6}}\ \mathrm{T}$

(C) $\sqrt{\frac{5}{4}}\ \mathrm{T}$

(D) $\sqrt{\frac{6}{5}}\ \mathrm{T}$

Q.14

A uniform rod of length L and mass M is pivoted at the centre. Its two ends are attached to two springs of equal spring constants k. The springs are fixed to rigid supports as shown in the figure, and the rod is free to oscillate in the horizontal plane. The rod is gently pushed through a small angle θ in one direction and released. The frequency of oscillation is



(A) $\frac{1}{2\pi}\sqrt{\frac{2k}{M}}$

(B) $\frac{1}{2\pi}\sqrt{\frac{k}{M}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{6k}{M}}$

(D) $\frac{1}{2\pi}\sqrt{\frac{24k}{M}}$

Q.15

A pendulum with time period of 1 s is losing energy due to damping. At certain time its energy is $45\ \mathrm{J}$. If after completing 15 oscillations, its energy has become $15\ \mathrm{J}$, then what will be the its damping constant (in $\mathrm{s^{-1}}$)?

(A) 0.0632

(B) 0.0532

(C) 0.0832

(D) 0.0732

Q.16

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The displacement of a particle in simple harmonic motion in one time period is $[\mathrm{A}=$ amplitude $]$

(A) A

(B) $2\ \mathrm{A}$

(C) $4\ \mathrm{A}$

(D) Zero

Q.17

Suppose a tunnel is dug along a diameter of the earth. A particle is dropped from a point, a distance h directly above the tunnel, the motion of the particle is

(A) simple harmonic

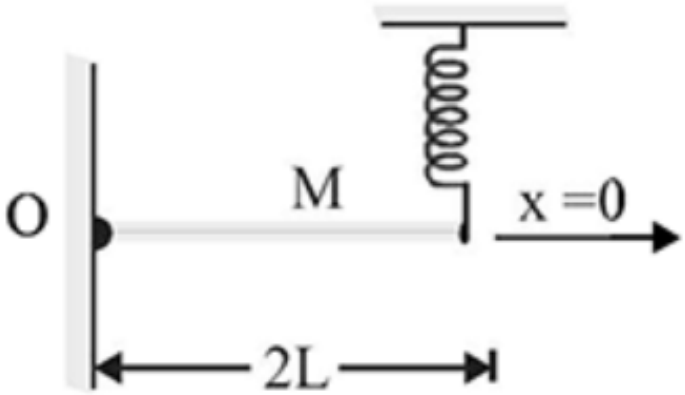
(B) parabolic

(C) oscillatory

(D) non-periodic

Q.18

A uniform pole of length $\ell=2\ \mathrm{L}$ is laid on smooth horizontal table as shown in figure. The mass of pole is M and it is O connected to a frictionless axis at O.A spring with force constant k is connected to the other end. The pole is displaced by a small angle θ_0 from equilibrium position and released such that it performs small oscillations. Then



(A) $\omega_0=\sqrt{\frac{\mathrm{M}}{3\mathrm{k}}}$

(B) $\omega_0=\sqrt{\frac{\mathrm{k}}{3\mathrm{M}}}$

(C) $\omega_0=\sqrt{\frac{3\mathrm{k}}{\mathrm{M}}}$

(D) $\omega_0=\sqrt{\frac{\mathrm{k}}{2\mathrm{M}}}$

Q.19

A rod of length ℓ is in motion such that its ends A and B are moving along x -axis and y -axis respectively. It

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is given that $\frac{d\theta}{dt} = 2\text{rad/sec}$ always. P is a fixed point on the rod. Let M be the projection of P on x -axis. For the time interval in which θ changes from 0 to $\frac{\pi}{2}$, the correct statement is

(A) The acceleration of M is always directed towards right

(B) M executes SHM

(C) M moves with constant speed

(D) M moves with constant acceleration

Q.20

A body executes simple harmonic motion under the action of a force F_1 with a time period $\frac{4}{5}$ s. If the force is changed to F_2, it executes S.H.M. with time period $\frac{3}{5}$ s. If both the forces F_1 and F_2 act simultaneously in the same direction on the body, then what will be the time period in second?

(A) 0.6

(B) 0.9

(C) 0.5

(D) 0.8

Answers & Solutions

Q.1 Answer:

$\omega_1 = \omega_2$

Solution:

In forced oscillations, the amplitude as well as the energy of the oscillator is maximum at resonance. Resonance occurs if the frequency of the applied force is equal to the frequency of the oscillator, i.e. when $\omega_1 = \omega_2 = \omega_0$. Hence the correct choice is (a).

Q.2 Answer:

$+6$ cm

Solution:

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The displacements of the particle at $x = 1$ cm at $t = 1$ s are given by

$$y_1 = 2\sin\left(3\pi \times 1 - \frac{\pi}{2} \times 1\right)$$
$$= 2\sin\left(3\pi - \frac{\pi}{2}\right)$$
$$= 2\sin\left(\frac{5\pi}{2}\right) = 2\sin\left(2\pi + \frac{\pi}{2}\right)$$
$$= 2\sin\frac{\pi}{2} = 2 \text{ cm}$$

and $y_2 = 4\sin\left(6\pi \times 1 - \frac{3\pi}{2} \times 1\right)$
$= 4\sin\left(\frac{9\pi}{2}\right) = 4\sin\left(4\pi + \frac{\pi}{2}\right)$
$= 4\sin\frac{\pi}{2} = 4$ cm

According to superposition principle

$y = y_1 + y_2 = 2 + 4 = 6$ cm

Q.3 Answer:

$2\sqrt{3}$ cm

Solution:

Resultant amplitude is $A = 2a\cos\left(\frac{\phi}{2}\right)$
$= 2 \times (2 \text{ cm}) \times \cos\left(\frac{60^\circ}{2}\right)$
$= 2\sqrt{3}$ cm

Q.4 Answer:

153 s

Solution:

Let m be the mass and l be the length of the stick.

$l = 100$ cm

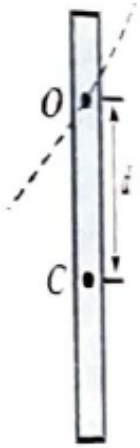
The distance of the point of suspension from centre of gravity is $d = 25$ cm

Moment of inertia about a horizontal axis through O is

$$I = I_C + md^2 \Rightarrow I = \frac{ml^2}{12} + md^2$$
$$\Rightarrow \quad T = 2\pi\sqrt{\frac{I}{mgd}} = 2\pi\sqrt{\frac{\frac{ml^2}{12} + md^2}{mgd}}$$

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$\Rightarrow \quad T = 2\pi\sqrt{\frac{I^2+12d^2}{12gd}} = 2\pi\sqrt{\frac{l^2+12(0.25)^2}{12\times 9.8\times 0.25}} = 153 \text{ s}$



Q.5 Answer:

1.26 s, 1.6 cm,6.8 cm/s, 1 rad/s^2

Solution:

(a) The angular frequency is $\omega = \sqrt{\frac{g}{l}} = \sqrt{\frac{10 \text{ m/s}^2}{0.4 \text{ m}}} = 5 \text{ s}^{-1}$

The time period is $\frac{2\pi}{\omega} = \frac{2\pi}{5s^{-1}} = 1.26 \text{ s}$

(b) Linear amplitude $= 40 \text{ cm} \times 0.04 = 1.6 \text{ cm}$

(c) Angular speed $\dot{\theta} = \omega\sqrt{\theta_0^2 - \theta^2}$

$\dot{\theta} = \left(5 \text{ s}^{-1}\right)\sqrt{(0.04)^2 - (0.02)^2}\text{rad} = 0.17\text{rad/s}$

where speed of the bob at this instant

$V = \dot{\theta}A = (40 \text{ cm}) \times (0.17 \text{ s}) = 6.8 \text{ cm/s}$

(d) At momentary rest, the bob is in extreme position. Thus, the angular acceleration $|\alpha| = \omega^2\theta$.

$|\alpha| = (0.04\text{rad})\left(25 \text{ s}^{-2}\right) = 1\text{rad/s}^2$

Q.6 Answer:

$\left(\frac{\pi}{2} + 1\right)$

Solution:

The block moving with kinetic energy $\frac{1}{2}mu^2$ **strikes the spring and compresses it until all the kinetic energy is converted into potential energy of the spring. Then the block moves to the left as spring begins to regain its original length** BC**. At this instant the block loses contact with the spring and begins to move towards wall** A **with velocity** u**. The time** t_1 **taken by the block to strike** C **and lose contact at** C **= half the time period of oscillation, i.e.**

$$t_1 = \frac{1}{2} \times 2\pi\sqrt{\frac{m}{k}}$$

$$= \pi\sqrt{\frac{2.5}{10}} = \frac{\pi}{2} \text{ second}$$

The block then travels towards A **with velocity** u **and rebounds with the same velocity** u **(since the collision is elastic) and agains strikes** C **after another time** t_2 **given by**

$t_2 = \frac{2x}{v} = \frac{2\times 0.5}{1} = 1 \text{ second}$

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This whole process is repeated. The block undergoes periodic motion with time period

$T = t_1 + t_2 = \left(\frac{\pi}{2} + 1\right)$ second

Q.7 Answer:

$\mathrm{mg}\, l(1 - \cos\theta)$

Solution:

When the bob moves from maximum angular displacement θ to mean position, then the loss of gravitational potential energy is mgh **where** $h = l(1 - \cos\theta)$

Q.8 Answer:

$2f$

Solution:

The two springs are in parallel. $\therefore$ Effective spring constant, $K = K_1 + K_2$ **Now, frequency of oscillation is given by**

$$f = \frac{1}{2\pi}\sqrt{\frac{K}{m}} \text{ or, } f = \frac{1}{2\pi}\sqrt{\frac{K_1 + K_2}{m}}$$

When both k_1 and k_2 are made four times their original values, the new frequency is given by

$$\begin{aligned} f' &= \frac{1}{2\pi}\sqrt{\frac{4K_1 + 4K_2}{m}} \\ &= \frac{1}{2\pi}\sqrt{\frac{4\left(K_1 + 4K_2\right)}{m}} = 2\left(\frac{1}{2\pi}\sqrt{\frac{K_1 + K_2}{m}}\right) \\ &= 2f; \quad \text{from eqn. (i)} \end{aligned}$$

Q.9 Answer:

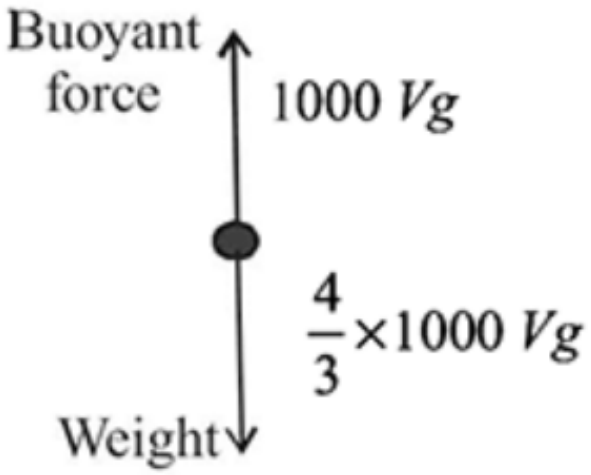
$t = 2t_0$

Solution:

$t = 2\pi\sqrt{\frac{\ell}{g_{\text{eff}}}}; t_0 = 2\pi\sqrt{\frac{\ell}{g}}$

$\left.\begin{array}{l}\text{Buoyant} \\ \text{force} \\ \text{Weight}\end{array}\right] \frac{4}{3} \times 1000\mathrm{Vg}$

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Net force $= \left(\frac{4}{3}-1\right)\times 1000\mathrm{Vg} = \frac{1000}{3}\mathrm{Vg}$

$g_{\text{eff}} = \frac{1000\mathrm{Vg}}{3\times\frac{4}{3}\times 1000\ \mathrm{V}} = \frac{g}{4}$

$\therefore t = 2\pi\sqrt{\frac{\ell}{g/4}}$

$t = 2t_0$

Q.10 Answer:

$\left[\left(\frac{\mathrm{T_M}}{\mathrm{T}}\right)^2 - 1\right]\frac{\mathrm{A}}{\mathrm{Mg}}$

Solution:

As we know, time period, $\mathrm{T} = 2\pi\sqrt{\frac{\ell}{\mathrm{g}}}$ **When additional mass** M **is added then** $\mathrm{T_M} = 2\pi\sqrt{\frac{\ell+\Delta\ell}{\mathrm{g}}}$

$\mathrm{T}_{\frac{\mathrm{M}}{\mathrm{T}}} = \sqrt{\frac{\ell+\Delta\ell}{\ell}}$ **or** $\left(\frac{\mathrm{T_M}}{\mathrm{T}}\right)^2 = \frac{\ell+\Delta\ell}{\ell}$

or, $\left(\frac{\mathrm{T_M}}{\mathrm{T}}\right)^2 = 1+\frac{\mathrm{Mg}}{\mathrm{AY}} \quad \left[\because \Delta\ell = \frac{\mathrm{Mg}\ell}{\mathrm{AY}}\right]$

$\therefore \frac{1}{\mathrm{Y}} = \left[\left(\frac{\mathrm{T_M}}{\mathrm{T}}\right)^2 - 1\right]\frac{\mathrm{A}}{\mathrm{Mg}}$

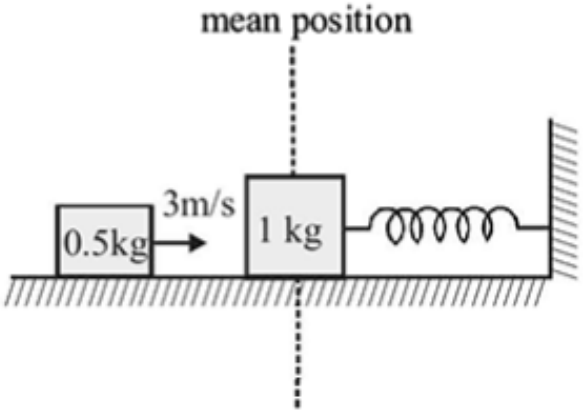
Q.11 Answer:

$5\ \mathrm{cm}, \frac{\pi}{10}\ \mathrm{s}$

Solution:

Applying linear momentum conservation, $0.5\times 3 = (1+0.5)\mathrm{v}$ **or** $\mathrm{v} = 1\ \mathrm{m/s}$

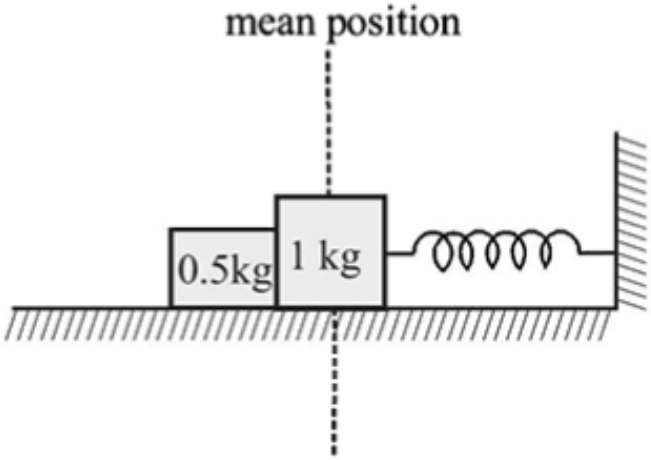
By conservation of energy,



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After collision

$\frac{1}{2}(1+0.5)v^2 = \frac{1}{2}kA^2 \Rightarrow A = \sqrt{\frac{1.5}{k}} \times v$



$\Rightarrow A = \sqrt{\frac{1.5}{600}} \times 1 = \frac{1}{20}\ m = 0.05\ m$

$A = 5\ cm.$

Time period of oscillation,

$T = 2\pi\sqrt{\frac{m_1+m_2}{k}} = 2\pi\sqrt{\frac{1.5}{600}} = \frac{2\pi}{20} = \frac{\pi}{10}\ s$

Q.12 Answer:

$\frac{mg}{k}\left[\sqrt{1+\frac{2kh}{mg}}-1\right]$

Solution:

Loss in PE of mass = gain in PE of the spring

$\therefore \quad mg(h+x) = \frac{1}{2}kx^2$

$kx^2 - 2mgx - 2mgh = 0$

$\Rightarrow x = \frac{2mg \pm \sqrt{4\,m^2\,g^2 - 4k(-2mgh)}}{2k}$

or $\quad x = \frac{mg}{k} \pm \sqrt{\frac{m^2\,g^2}{k^2} + \frac{2mgh}{k}}$

$$x = A - \frac{mg}{k} = \sqrt{\frac{m^2\,g^2}{k^2} + \frac{2mgh}{k}} - \frac{mg}{k}$$

$$= \frac{mg}{k}\left[\sqrt{1+\frac{2kh}{mg}} - 1\right]$$

Q.13 Answer:

$\sqrt{\frac{5}{6}}\ T$

Solution:

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Distance covered by lift is given by $y = t^2$

$\therefore$ **Acceleration of lift upwards** $= \frac{d^2y}{dt^2} = \frac{d}{dt}(2t) = 2\ m/s^2 = \frac{g}{5}$

Now, $T = 2\pi\sqrt{\frac{\ell}{g}}$

$T' = 2\pi\sqrt{\frac{\ell}{g+\frac{g}{5}}} = 2\pi\sqrt{\frac{\ell}{\frac{6}{5}g}} = \sqrt{\frac{5}{6}}\,T$

Q.14 Answer:

$\frac{1}{2\pi}\sqrt{\frac{6k}{M}}$

Solution:

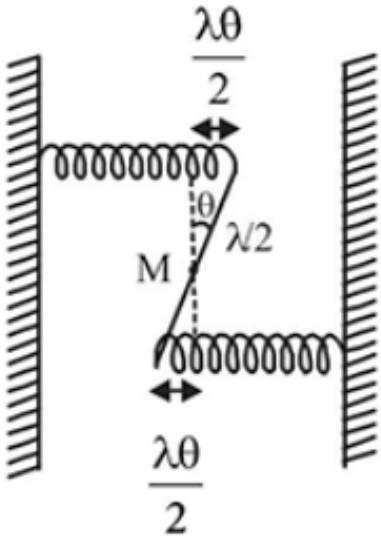


Figure shows the rod at an angle θ **with respect to its equilibrium position. Both the springs are stretched by length** $\frac{\ell\theta}{2}$ M

The restoring torque due to the foopoooopoo springs

$\tau = -2$ **(Restoring force)** $\times$ 目 $\frac{\lambda\theta}{2}$

perpendicular distance $\tau = -2k\left(\frac{\ell\theta}{2}\right) \times \frac{\ell}{2} = -k\frac{\ell^2}{2}\theta$ **.....(i)**

If I **is the moment of inertia of the rod about** M **then**

$$\tau = I\alpha = I\frac{d^2\theta}{dt^2}$$

.....(ii)

From (i) \& (ii) we get $I\frac{d^2\theta}{dt^2} = -k\frac{\ell^2\theta}{2} \Rightarrow \frac{d^2\theta}{dt^2} = -\frac{k}{I}\frac{\ell^2}{2}\theta = \frac{-k}{M\ell^2/12}\frac{\ell^2}{2}\theta$

$\Rightarrow \frac{d^2\theta}{dt^2} = -\frac{6k}{M}\theta$

Comparing it with the standard equation of rotational SHM we get

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$$\frac{d^2\theta}{dt^2} = -\omega^2\theta \Rightarrow \omega^2 = \frac{6k}{M} \Rightarrow \omega = \sqrt{\frac{6k}{M}}$$

$$\Rightarrow 2\pi v = \sqrt{\frac{6k}{M}} \Rightarrow v = \frac{1}{2\pi}\sqrt{\frac{6k}{M}}$$

Q.15 Answer:

0.0732

Solution:

As we know, $E = E_0 e^{-\frac{bt}{m}}$ $15 = 45e^{-\frac{b15}{m}}$

[As no. of oscillations $= 15$ **so** $t = 15\text{sec}$ **]**

$\frac{1}{3} = e^{-\frac{b15}{m}}$

Taking log on both sides

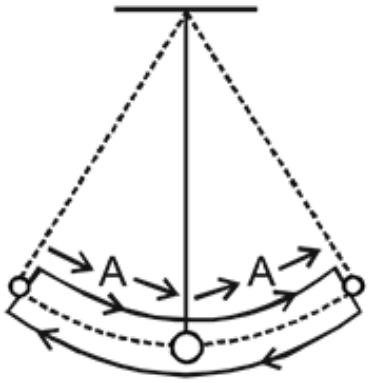
$\frac{b}{m} = \frac{1}{15}\ell\text{n}3 = 0.0732$

Q.16 Answer:

Zero

Solution:

As seen from figure after one time period the bob return to its equilibrium position, so diplacement of the particle is zero, but distance covered by the particle in one time period is 4 A **(where** A **is amplitude of bob, when it does S.H.M.)**

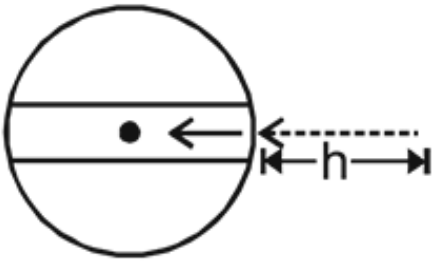


Q.17 Answer:

oscillatory

Solution:

When a particle is dropped from a height h **above the centre of tunnel.**



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(i) It will oscillate, through the earth to a height h on both sides.

(ii) The motion of particle is periodic.

(iii) The motion of particle will not be S.H.M.

Q.18 Answer:

$\omega_0 = \sqrt{\frac{3\mathrm{k}}{\mathrm{M}}}$

Solution:

$\mathrm{I} = \frac{1}{3}\mathrm{M}(2\,\mathrm{L})^2 = \frac{4}{3}\mathrm{ML}^2$

Force applied by the spring is $\mathrm{F} = -\mathrm{kx} \Rightarrow \mathrm{F} = -\mathrm{k}(2\,\mathrm{L}\theta)$

(θ is the angular displacement from the equilibrium position). Further

$\tau = |\vec{\ell} \times \overrightarrow{\mathrm{F}}| = 4\,\mathrm{L}^2\mathrm{k}\sin\theta = -4\,\mathrm{L}^2\mathrm{k}\theta$

Also, $\tau = I\alpha = I\ddot{\theta} = -4\,\mathrm{L}^2\mathrm{k}\theta$

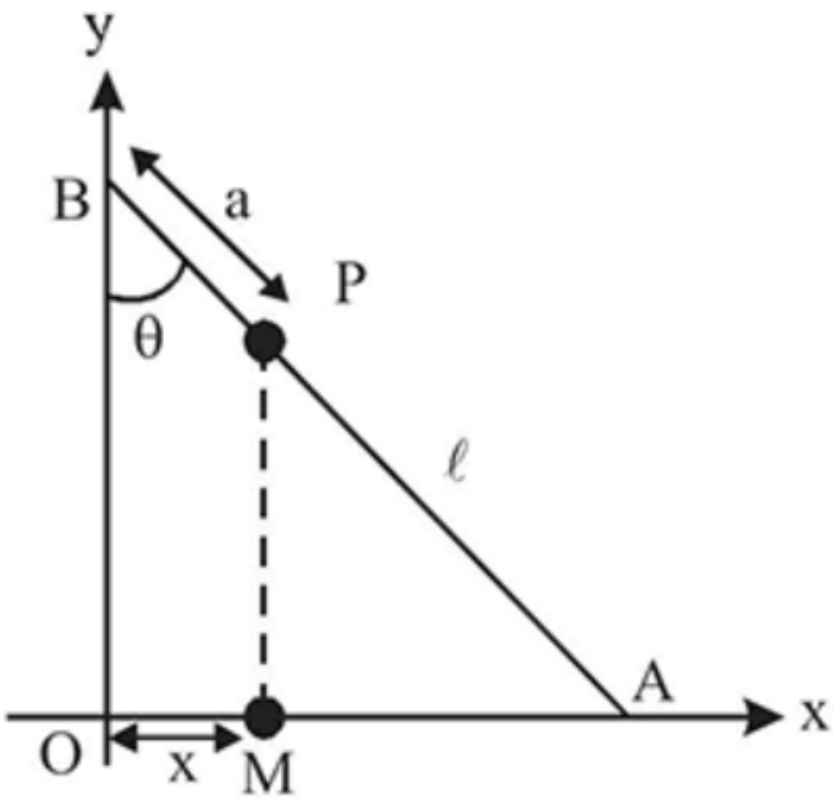
$\Rightarrow \ddot{\theta} + \frac{3\mathrm{k}}{\mathrm{M}}\theta = 0$

$\Rightarrow \omega_0 = \sqrt{\frac{3\mathrm{k}}{\mathrm{M}}}$

Q.19 Answer:

M executes SHM

Solution:



$\frac{\mathrm{d}\theta}{\mathrm{dt}} = 2 \therefore \theta = 2\mathrm{t}$

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Let $\mathrm{BP}=\mathrm{a}, \quad \therefore \mathrm{x}=\mathrm{OM}=\mathrm{a} \sin \theta=\mathrm{a} \sin (2 \mathrm{t})$

Hence M executes SHM within the given time period and its acceleration is opposite to x that means towards left.

Q.20 Answer:

0.8

Solution:

Under the action of first force, $\mathrm{F}_1=\mathrm{m} \omega_1^2 \mathrm{y}$ Under the action of second force,

$\mathrm{F}_2=\mathrm{m} \omega_2^2 \mathrm{y}$

Under the action of resultant force, $\mathrm{F}_1+\mathrm{F}_2=\mathrm{m} \omega^2 \mathrm{y}$

$\Rightarrow m \omega^2 y=m \omega_1^2 y+m \omega_2^2 y$

$\Rightarrow \omega^2=\omega_1^2+\omega_2^2$

$\Rightarrow\left(\frac{2 \pi}{\mathrm{T}}\right)^2=\left(\frac{2 \pi}{\mathrm{T}_1}\right)^2+\left(\frac{2 \pi}{\mathrm{T}_2}\right)^2$

$\Rightarrow T=\sqrt{\frac{T_1^2 T_2^2}{T_1^2+T_2^2}}=\sqrt{\frac{\left(\frac{4}{5}\right)^2 \cdot\left(\frac{3}{5}\right)^2}{\left(\frac{4}{5}\right)^2+\left(\frac{3}{5}\right)^2}}=\frac{12}{25}$.

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